Physiology & Behavior, Vol. 38, pp. 795–801. Copyright © Pergamon Journals Ltd., 1986. Printed in the U.S.A. 0031-9384/86 $3.00 + .00

Contrasting Responses to Intruders and to Involuntary Separation by Monogamous and Polygynous New World Monkeys

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Received 19 May 1986

MENDOZA, S. P. AND W. A. MASON. *Contrasting responses to intruders and to involuntary separation by monogamous and polygynous New World monkeys.* PHYSIOL BEHAV 38(6) 795–801, 1986.—The strength and quality of the relationship between heterosexual pairmates were compared in two species of New World monkeys, the polygynous squirrel monkey (*Saimiri*) and the monogamous titi monkey (*Callicebus*). This was accomplished by measuring behavioral and adrenocortical responses to intruders and to involuntary separation from the pairmate. The species diverged sharply in their responses to these conditions. For titi monkeys, the presence of an intruder resulted in a marked increase in behavioral signs of agitation, which were more prominent in the subject of the same sex as the intruder. Most male titi monkeys showed an increase in plasma cortisol in the presence of a male intruder; this pattern was not seen in their response to a female intruder. Female titi monkeys showed no consistent pattern of adrenocortical response to intruders of either sex. For squirrel monkeys, the presence of an intruder resulted in increased frequency of displays in male subjects, but not in female subjects. Both sexes showed a reliable reduction in plasma cortisol, below basal levels, in the presence of a female intruder, whereas no effect on cortisol was produced by a male intruder. Titi monkeys, but not squirrel monkeys, responded to separation with a significant increase in plasma cortisol and in behavioral indices of agitation or distress.

Social relationships Pituitary-adrenal activity Monogamy Polygyny *Callicebus* *Saimiri*

IN primate societies the same individuals tend to interact with each other repeatedly and in more or less predictable ways. It is this aspect of primate behavior which led Hinde and his associates to emphasize long-term relationships as critical elements in the description of social organization [24–27]. Primate species show considerable variation in the types and qualities of relationships in which individuals engage and, consequently, in the organization of their social groups. For each species, however, there is a readily identifiable modal grouping pattern which is maintained despite changes in the members of the group [33] and which is usually generated rapidly following formation of new social units in the laboratory [29, 35, 36].

The formation and maintenance of species-characteristic forms of social organization implies processes whereby certain types of relationships or associations are facilitated, while others are inhibited or excluded. For example, maintenance of a monogamous social structure is presumably based on the affinitive relationship between an adult male and an adult female and the exclusion of extraneous adults. Similar processes probably occur in the organization of multi-male, multi-female groups. For example, in established social groups of macaques, a predictable response to the introduction of unfamiliar individuals is intense and prolonged aggression directed toward the strangers [5,43]; when groups are formed by introducing individuals who are all unfamiliar with one another, however, aggression is generally brief and less intense, or absent altogether [4,35].

These findings could be viewed as supporting the general hypothesis that the form and intensity of the reaction to strangers is influenced by the presence or quality of existing social relationships. When groups are formed *de novo*, or when existing relationships are unsatisfactory, strangers may offer the potential for developing new relationships. Alternatively, in groups in which established relationships are important and satisfying, strangers may pose a threat to their integrity and be the target of attack or other exclusionary behavior. From this standpoint, the response to unfamiliar conspecifics can be used to evaluate the status of existing relationships.

Another means of evaluating the quality or strength of a social relationship is to measure the response of the participants to disrupting or severing the relationship. This approach has most often employed involuntary separation, and has been used mainly in research on dyadic relationships, particularly that between mother and offspring. It has also

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been used effectively, however, in evaluating the quality, strength and specificity of relationships between nonkin, including juveniles and adults (e.g., [22,39]).

The present study used both of these approaches to compare adult heterosexual relationships in two species of South American monkeys, *Saimiri sciureus* (squirrel monkeys) and *Callicebus moloch* (titi monkeys). All subjects were living as established heterosexual pairs. In nature, however, titi monkeys are found in family groups typically consisting of an adult pair and one or two offspring, whereas squirrel monkeys live in large mixed-sex groups in which males and females associate preferentially with members of their own sex [2, 9, 28, 30, 31, 34, 44, 45]. In this research pairmates of both species were confronted in their living cages with a single male or female conspecific presented behind a clear plastic screen, and with involuntary separation from the pairmate. In addition to monitoring behavioral reactions to these conditions, adrenocortical responses were evaluated. The adrenocortical system was selected because previous research on various species has established that it provides a sensitive index of psychological responses to environmental events.

In previous research we have demonstrated that adrenocortical activity is consistently higher in squirrel monkeys than in titi monkeys [12,41]. Our aim in the present study was to assess the relative responsiveness of this system in the two species to the experimental conditions. Their adrenocortical response to involuntary separation from pairmates has not been compared previously. The adrenocortical response to strangers was examined in one experiment in which the monkeys were tested in a novel environment. Although no evidence was obtained of an adrenocortical response to strangers in either species, the effect of the novel environment was substantial and could easily have masked the response to social variables [1]. For this reason, the present research was carried out with the monkeys in their familiar living cage.

Based on the differing life styles of the two species, and on the available experimental data, we predicted that the adrenocortical response to the test conditions would be more vigorous in titi monkeys [1, 10, 18, 22, 32]. Although we also considered it likely that the sexes would differ within each species in some aspects of their adrenocortical responses to the experimental conditions, the precise pattern of differences could not be predicted from the available data.

METHOD

Subjects and Living Arrangements

Ten adult heterosexual pairs of *Callicebus* and ten adult heterosexual pairs of *Saimiri* served as subjects. Seven *Callicebus* (5 males, 2 females) and 17 *Saimiri* (9 males, 8 females) were wild-caught. Specific pairing arrangements had been in effect for a mean of 39.3 months for *Callicebus* (range: 2 to 113) and 41.6 months for *Saimiri* (range: 3 to 110). Five *Callicebus* pairs lived with subadult offspring. Each pair was housed in an indoor-outdoor cage (1×1×3 m indoor portion; 4×1×2 m outdoor portion). The indoor portion of the living cage contained a standard arrangement of perches and was separated from the outdoor portion by an opaque wall containing a doorway (20×30 cm) which the monkeys could approach from the inside by a single perch. All animals were maintained according to standard laboratory protocol, which included cleaning at 0800 hr and feeding at 1300 hr.

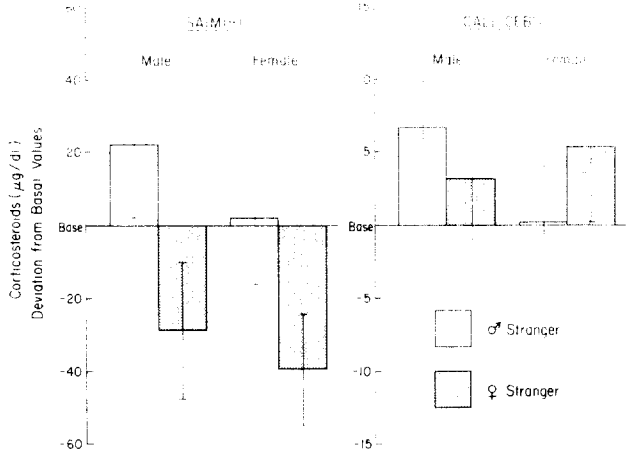


FIG. 1. Corticosteroid deviations (mean±SE) from basal values following one-hour exposure to unfamiliar male and female conspecifics. Note: Results are presented in different scales to facilitate comparisons within and between species.

Procedures

On the evening before testing, subadult offspring were removed and adult pairs were confined to the indoor portion of the living cage. This was accomplished by blocking the doorway between the inside and the outside portions with a small stimulus-cage having a clear plastic front. A guillotine door at the rear of this cage permitted stimulus-animals to be introduced from outside with minimal disturbance to the subjects. To further minimize disturbance, all subjects had been previously trained to enter a small transport cage. Testing was conducted between 1500 and 1800 hr.

Blood samples for determination of plasma corticosteroids were collected from each subject in each of four conditions:

(1) Base: One member of the pair was caught and a blood sample was collected for determination of resting corticoid levels. Base samples were obtained on separate days for each member of a pair.

(2) Disturbance: This condition occurred in conjunction with the Base condition and was designed to assess the effects of the procedures used to capture and remove one animal for the Separation condition on the animal remaining in its living cage. Following collection of the basal sample, the captured subject was returned to the home cage and its pairmate. One hour later the pairmate was caught and its blood was sampled. This procedure was performed once to determine disturbance effects on the female, and once to determine disturbance effects on the male.

(3) Stranger: A single unfamiliar conspecific was placed in the stimulus-cage attached to the living cage. One hour later both subjects were caught and their blood was sampled. Each pair was tested twice, once with an unfamiliar adult male, and once with an unfamiliar adult female.

(4) Mate Separation: One member of the pair was caught and removed from the living area (and out of auditory range) with as little disturbance as possible to the pairmate. One hour following separation, the subject remaining in the home cage was caught and its blood was sampled. This condition was imposed twice for each pair, once to evaluate the male's response to separation and once to evaluate the female's response to separation.

TABLE 1
BEHAVIORAL RESPONSES TO STRANGERS

		Saimiri			*Callicebus*		
		BASE	♂ STR	♀ STR	BASE	♂ STR	♀ STR
Movement*	♂	18.7	18.0	19.7	6.1	12.7	13.2
	♀	19.6	19.2	22.3	6.4	13.2	10.9
Vocalization*	♂	2.9	6.7	4.4	0.5	5.8	4.4
	♀	2.4	2.5	1.9	1.0	6.8	7.7
Approach STR†	♂	7.8	12.9	17.3	6.0	14.4	15.3
	♀	5.6	2.8	4.8	3.7	9.8	5.4
Display†	♂	0.7	13.4	8.0	0.5	26.7	16.1
	♀	0.0	0.0	1.4	0.1	5.7	14.9
Approach Mate†	♂	2.6	3.5	1.9	6.7	19.0	14.2
	♀	8.4	8.4	5.9	5.6	12.5	12.8
Pair Contact*		2.0	2.7	2.0	19.8	16.1	13.7
Pair Proximity*		5.1	7.6	5.0	23.9	25.3	23.4

*Number of 15-sec intervals (max=40).
†Frequency of occurrence/40 min.

Behavioral data were collected during conditions 2–4 above. In addition, behavioral observations were made on a day when no blood samples were collected, but all other procedures were followed, including removing any subadult offspring and attaching the stimulus-cage on the evening before testing. This constitutes the base condition for behavioral comparisons.

Each pair was tested weekly. Only one pair in each caging area was tested on a given day, so that any disturbance occasioned by testing one pair would not influence results obtained for another. The order in which conditions were presented was approximately balanced across pairs and species.

Behavioral observations. Three types of behavioral data were collected during each 1-hr observation period and included measures of individual behavior, and (where appropriate) social distance and social interactions. Individual behaviors included vocalizations, locomotor activity, manipulation of objects (as indices of distress or arousal; see [22]) and behaviors which are believed to serve a scent-marking function in one or both species (rump rub, back rub, chest rub, urine wash, sneeze; see [19, 21, 42]). Individual behaviors were scored by 15-sec intervals (once per interval) in five 2-min sessions separated by ten minutes, yielding a maximum score of 40 per condition.

Measures of social distance between cagemates were obtained concurrently with individual behaviors and included contact and proximity (within arm's reach) for both species and tail-twine for *Callicebus*. Occurrences of these measures were scored once on-the-signal for each 15-sec interval.

Social interactions were scored during four 10-min sessions and included displays (genital display and cage-shake for *Saimiri*, lipsmacking and arching for *Callicebus*), approach, grab, active displace, passive displace, ano-genital exploration and social exploration (see [38,42]). Direct frequency counts were used for all social interactions. All interactions occurring between cagemates were recorded, along with the identity of the initiator and of the recipient. Approach and display behaviors directed to the stimulus-cage (whether a stranger was present or not) were scored separately.

Observations were conducted by experienced observers from behind a one-way viewing screen. Inter-observer reliabilities exceeded 90% agreement for all behavioral categories, based on 10 sessions for each recording procedure.

Blood sampling and cortisol determination. For blood sampling, subjects were captured in the transport cage, removed, and manually restrained while a blood sample (2 ml) was collected via femoral puncture. Samples were collected within a pre-determined limit of three minutes from the start of the capture procedure. For the stranger conditions, blood samples from both members of the test pair were obtained within the same 3-min period. Studies with rats indicate that corticosteroid levels for blood samples collected within this time limit are not influenced by the capture and handling procedure [13]. Blood samples were centrifuged and the plasma fraction extracted and frozen (−20°C) until assayed by a competitive protein binding technique adapted from that of Bassett and Hinks [3]. All samples were analyzed in duplicate in a single assay. Intra-assay coefficient of variation was 6.0%. Although this assay measures both cortisol and corticosterone concentrations, comparison of values obtained by this method with those obtained by radioimmune assay procedures specifically for cortisol do not yield significant differences; hence, results are presented with reference to cortisol. Two *Callicebus* were determined to have been in the early stages of pregnancy during the study. Their data were not included in the hormonal analysis.

Data analysis. Data were analyzed separately to compare each species' response to strangers and their response to mate separation. Cortisol values were evaluated by analysis of variance, with species and sex of subject treated as between group factors, and conditions treated as a repeated measure. Behavioral measures were analyzed nonparametri-

TABLE 2
BEHAVIORAL RESPONSES TO SEPARATION

		Saimiri			*Callicebus*		
		BASE	DIST	SEP	BASE	DIST	SEP
Movement*	♂	18.7	22.0	22.8	6.1	6.9	12.7
	♀	19.6	28.4	23.9	6.4	12.1	14.3
Vocalization*	♂	2.9	3.0	6.6	0.5	2.6	5.9
	♀	2.4	4.7	4.6	1.0	5.2	11.8

*Number of 15-sec intervals (max=40).

cally using the Friedman test for overall comparisons among conditions, the Wilcoxon signed-ranks test for pairwise comparisons, and the Mann-Whitney test for comparisons between sexes and species. Correlational analyses were performed using Pearson's product-moment statistic.

RESULTS

Adrenocortical Response to Strangers

Analysis of basal plasma cortisol levels and response to the presence of strangers indicated that *Saimiri* showed substantially higher adrenocortical activity overall than did *Callicebus* (*Saimiri*: mean=186.3 μg/dl; *Callicebus*: mean=30.3 μg/dl; $F(1,34)=70.23, p<0.001$), and that the two species showed divergent patterns of response to the strangers, $F(2,68)=6.12$, $p<0.01$ (see Fig. 1). Mean values indicated that titi monkeys tended to respond to like-sex strangers with a modest elevation in plasma cortisol. This effect was more pronounced in males; 8 of 10 males responded to the male stranger with an elevation in corticosteroid levels, as compared to basal values, whereas only 4 of 8 females responded similarly to a female stranger. As a group, neither males nor females showed a consistent pattern in their adrenocortical responses to a stranger of the opposite sex. Separate analysis of the adrenocortical data for titi monkeys indicated that the effect of strangers was not significant. In contrast, male and female squirrel monkeys both responded to strangers. This was reflected in a significant main effect for conditions when their data were analyzed separately from those of the titi monkeys, $F(2,36)=6.63$, $p<0.01$. Surprisingly, this outcome could be attributed to a reduction, below basal levels, following presentation of a female stranger (see Fig. 1). This effect was apparent in 8 of 10 males and 7 of 10 females. Presence of a male stranger had no consistent effect on cortisol levels in subjects of either sex. Post-hoc comparisons revealed that corticoid values were significantly lower for the Female Stranger condition than were those for either the Base or Male Stranger conditions (p's<0.05).

Behavioral Response to Strangers

In the presence of strangers, male squirrel monkeys characteristically moved into proximity to the stimulus-animal and either remained in that position for the duration of the test, or made frequent approaches to and withdrawals from the stranger. A given male tended to be consistent in this respect with male and female strangers, and males were evenly divided in the pattern they exhibited. Although mean values indicate that approaches to the stimulus-cage increased when a stranger was present, this effect was not significant (see Table 1). All males displayed on at least one occasion and displays occurred significantly more often during stranger presentations (Base: mean=0.7; Stranger: mean=10.7; $p=0.01$); displays were more frequent in the presence of male than of female strangers (mean=13.4 vs. 8.0, respectively), but the difference was not significant ($p<0.10$). Females generally remained well away from strangers, and rarely displayed. For both sexes, individual behaviors and behaviors directed to the mate were uninfluenced by presence of strangers, regardless of gender.

For *Callicebus*, presence of strangers altered individual behaviors and the interactions between cagemates; these effects were accompanied by behavioral reactions directed toward the stranger (see Table 1). Movement increased in both sexes when a stranger was present (Base: mean=6.2; Male Stranger: mean=12.9; Female Stranger: mean=12.0; $p=0.001$), as did the frequency of vocalizations (Base: mean=0.7; Male Stranger: mean=6.3; Female Stranger: mean=6.0; $p<0.01$). Typically, *Callicebus* moved back and forth between the mate and the stranger, resulting in significant increases in approaches to the mate (Base: mean=6.1; Male Stranger: mean=15.7; Female Stranger: mean=13.5; $p<0.01$) and to the stimulus-cage (Base: mean=4.8; Male Stranger: mean=12.1; Female Stranger: mean=10.3; $p<0.001$). Observations during the Base condition indicated that titi monkey pairmates were within proximity to one another on 60% of the intervals. Despite the increase in activity in the presence of strangers, contact and proximity scores were not reduced as compared to the Base condition, suggesting that a focus on the pairmate was maintained. Displays occurred predominantly when strangers were present (Base: mean=0.3; Male Stranger: mean=16.2; Female Stranger: mean=15.5; $p<0.01$), and were exhibited more frequently by the subject of the same sex as the stranger (same sex: mean=20.8; opposite sex: mean=10.9; $p<0.05$). Displays in the presence of same-sex intruders were more frequent in males than in females (mean=26.7 vs. 14.9, respectively). In contrast to squirrel monkeys, *Callicebus* directed their displays equally often to the mate and stranger, although they differentiated between them in type of display. Arching was directed to the stranger significantly more often than to the mate (Stranger: mean=6.8; Pairmate: mean=2.6; $p<0.05$), whereas lipsmacking tended to be directed more often to the mate than to the stranger (Stranger: mean=1.3; Pairmate: mean=2.4; NS). A frequent observation was that both males and females interposed themselves between their cagemate and a like-sex stranger, displaying alternately to the mate and to the stranger.

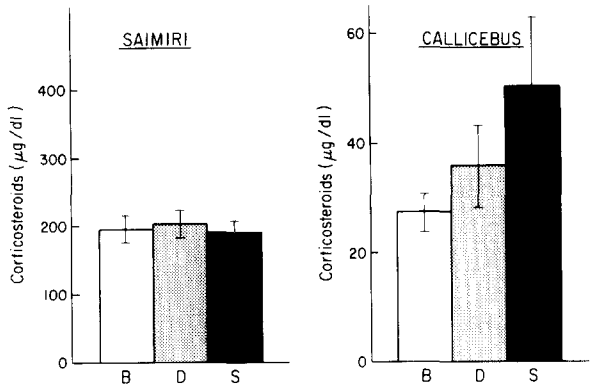


FIG. 2. Mean corticosteroid values (±SE). B=base condition; D=disturbance control; S=following one-hour separation from cagemate. Note: Results are presented in different scales.

Response to Separation

Squirrel monkeys did not respond to separation from their cagemates, as measured by behavior or physiology. Values for plasma cortisol, movement, vocalization, scent-marking and object manipulation were unaffected by either the Disturbance or Mate Separation conditions. In contrast, *Callicebus* responded to separation with reliable changes in behavior and physiology. Frequencies of vocalization and movement were higher while subjects were separated from their cagemates than during either the Base or Disturbance conditions (Vocalization, Base: mean=0.7; Disturbance: mean=3.9; Separation: mean=8.8; Movement, Base: mean=6.2; Disturbance: mean=9.5; Separation: mean=13.5; p's<0.05, see Table 2). Plasma cortisol levels for *Callicebus* also differentiated conditions, $F(2,32)=3.80$, $p<0.05$ (see Fig. 2), and attained their highest mean values during Mate Separation. Examination of individual values indicated that at least part of the effect could be attributed to the subject's response to capture of its mate. Cortisol levels for 12 of 18 titi monkeys were higher following Disturbance, as compared to the Base condition; 13 of 18 titis also showed higher levels following Separation than following Disturbance. This pattern of response resulted in separation values that were significantly elevated over basal values ($p<0.05$), but no other pairwise comparison was significant.

Correlational Analyses

To compare the use of intruders and involuntary separation as methods for investigating social relationships, we examined the data for patterns of association between cortisol and other measures, across situations. The results for *Callicebus* suggest that, in spite of the finding that intruders and separation both resulted in an increase in mean cortisol values, these conditions were assessing different aspects of the relationship between pairmates. The correlation between the cortisol values in response to Disturbance and Separation was .84 ($p<0.001$), indicating a high degree of consistency in individual responsiveness to these procedures. In contrast, the correlations between cortisol responses to either Disturbance or Separation and the Stranger conditions were slight (range=.03 to .24), suggesting little, if any, individual consistency in response to these conditions. The cortisol responses to the intruder of the same sex and the intruder of the opposite sex, however, were positively correlated, ($r=.45$, $p<0.05$); frequency of displays to the intruder of the same sex and the intruder of the opposite sex were also positively correlated ($r=.74$, $p<0.001$). There was also a clear suggestion of a consistent association in males between cortisol values and the frequency of displays (same sex, $r=.64$, $p<0.05$; opposite sex, $r=.40$, NS); for females both coefficients were less than .20.

In contrast, cortisol values for squirrel monkeys were highly intercorrelated among all conditions (range=.62 to .83, p's<0.01), and gave no indication of differentiating between the intruder and separation conditions. Furthermore, although most animals showed a reduction in cortisol in the presence of the female stranger, the magnitude of this effect was consistent with the overall pattern of individual differences, as evidenced by a coefficient of .81 ($p<0.001$) between the responses to male and female intruders.

DISCUSSION

It is clear from these results that the titi monkey and the squirrel monkey—which are, respectively, monogamous and polygynous, in the wild—differ sharply in their responses to strangers and to involuntary separation, even when the species are housed and tested under identical conditions. For titi monkeys, separation resulted in an increase in plasma cortisol and in behavioral indices of agitation or distress. The presence of a stranger produced an increase in displays, modified the frequency of individual behavior, and changed the patterns of interaction between pairmates. Although the arousing effects of separation and of the intruders may be viewed as contributing to the preservation of a pair bond, correlational analyses of cortisol responses suggested that these two procedures were reflecting different aspects of the relationship between pairmates. The magnitude of the corticosteroid response to separation was not predictive of the response to strangers; on the other hand, the corticosteroid response to a stranger of one sex was significantly correlated with that to a stranger of the opposite sex. In general, males tended to show a more vigorous response to a stranger of the same sex than did females, as reflected in both behavior and physiology. This contrast between the sexes complements findings from other studies of more intense 'jealousy' in males [1,11]. In titi monkeys, the male is heavily involved in the transport of young [39]; on evolutionary grounds one might argue that the male's more vigorous behavioral and physiological reaction to same-sex intruders is part of an adaptive pattern that reduces the likelihood that he will be cuckolded and invest his parental efforts in caring for offspring that are not his own.

Squirrel monkeys, in contrast to titi monkeys, showed no behavioral or physiological response to separation from the pairmate, nor to the Disturbance condition. The presence of strangers had measurable effects on behavior and physiology, but these were attributable entirely to reactions to the stranger, and seemed to be independent of any existing relationship between pairmates. In particular, in the presence of a stranger, males tended to increase their approaches to the stimulus-cage (more with the female than with the male stranger) and the frequency of their displays (more to the male than to the female); females seldom approached the stimulus-cage or displayed, regardless of the sex of the stranger. Strangers had no effect on patterns of interaction between pairmates.

In spite of the clear behavioral indications of arousal in males, and the seeming indifference of the females to the

strangers, the majority of squirrel monkeys showed a reduction in plasma cortisol below basal values in the presence of a female stranger. We have no ready explanation for this surprising finding. Several psychological processes, including formation of coping strategies, engagement in consummatory or displacement behaviors (in the ethological sense), and interaction with attachment figures, have been shown to suppress the pituitary-adrenal response to arousing or stressful circumstances [6, 7, 20, 23, 37, 47, 48]. None of these processes were discernible in our experimental situation, and it seems unlikely that they contributed to the findings we obtained. Furthermore, in contrast to these studies, our results show a reduction below basal values, a relatively rare finding in psychoendocrine research. In the few instances where such a reduction has been found the effect has been linked to the anticipation or occurrence of pleasurable states or events [8, 14–16]. Our data clearly indicate that such an effect can be caused directly by social stimuli; moreover, it was obtained within one hour and in circumstances which obviated physical contact with the stimulus-animal.

It is intriguing that the reduction in adrenocortical activity in squirrel monkeys occurred only with the female stranger. We suspect that the fact that the monkeys were housed in heterosexual pairs (an arrangement which is atypical for this species) may have had an important influence, particularly on females. It is as though a female living with only a single male companion is experiencing a form of social deprivation, which an unfamiliar female promises to alleviate. In keeping with this suggestion, female squirrel monkeys housed with a single female companion show lower basal cortisol values than females living alone [17]. Moreover, females who are housed with only a single male show a strong attraction to unfamiliar females, as measured by proximity or by choice tests, and this effect is attenuated in females who are housed in groups with other females [32,46]. It may also be the case, however, as Hennessy's data [18] suggest, that an unfamiliar female is a particularly potent stimulus to another squirrel monkey female, regardless of her current living arrangements. Although the spatial data did not indicate strong female attraction to another female in the present results, we believe that this reflects the tendency of males to preempt the area in front of the stimulus-cage.

The reliable reduction in male cortisol levels in the presence of a female stranger is puzzling and difficult to reconcile with behavioral and physiological data from other studies. One might argue that males of a polygynous species who are living with a single female are, like their female pairmates, deprived of the normal measure of female companionship; other findings, however, provide no support for this interpretation. As a group, male squirrel monkeys, whether housed in heterosexual pairs or in mixed-sex social groups, do not show a predictable pattern of attraction to unfamiliar companions based on gender. Individual males, however, may show considerable consistency in their preferences for one sex or the other, or in the tendency to avoid strangers [32,46]. With respect to previous research on the pituitary-adrenal response of males to female strangers, a recurrent finding has been that their cortisol values increase over basal levels following introduction of females, rather than the reduction we found. Elevated adrenocortical activity in males has been found to accompany the formation of pairs, one-male groups, and multi-male groups [36,40]. In contrast to the present situation, however, the males in these studies were in direct physical contact with females. It is possible that our mode of presentation, which effectively prevented the stimulus-female from 'quitting the field' or otherwise frustrating the male's approaches, may have contributed to the effect we obtained.

The broad question that is our primary concern in this research and in our other comparative studies of *Saimiri* and *Callicebus* is: What factors contribute to the contrasting life styles of these two species? We have hypothesized that part of the answer will be found in the activity of physiological integrative systems, including the pituitary-adrenal system [1,12]. Our results are consistent with this expectation. The differences between species in the responses of the adrenocortical system to events that the animals could perceive as threatening the integrity of an established social relationship *(Callicebus)*, or offering the possibility of forming a new one *(Saimiri)*, are concordant with the life history data, but could not be predicted in detail from such information. Under natural conditions, of course, social systems are influenced by many variables, including accidents of demography, specific social traditions, and particular features of the current ecology [33]. Although such factors surely interact in complex ways to influence social relationships and social organization, they do not explain the recurrence of modal grouping patterns, across generations, in different populations, and in diverse environments. The principal significance of our findings is the light they shed on the organismic processes whereby individuals, through their interactions with each other, may produce and maintain the social systems characteristic of their species.

ACKNOWLEDGEMENTS

This research was suppored by Grant RR00169 from the National Institutes of Health, Division of Research Resources and National Research Service Award 1-F32-HD06127 to S. P. Mendoza. We thank E. N. Sassenrath for providing the cortisol assays and M. B. Hennessy for his helpful comments.

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